Faculty of Computers and Information Technology (FCIT)
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Bachelor of Computer Engineering

A System for Early Detection of Gas Leaks and Fires with IoT Alerts



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Academic Year: 2025/2026

1) ABSTRACT

Gas leaks and fire outbreaks are among the most dangerous hazards in residential and small industrial environments. This project presents an affordable and reliable early detection system that combines gas, smoke, and temperature sensing with local alarms and IoT-based smartphone notifications. Using an ESP32, MQ-2 sensor, DHT22 sensor, OLED display, LEDs, buzzer, and Blynk platform, the system continuously monitors the environment and classifies conditions into Normal, Warning, and Danger states. When hazardous conditions are detected, the system activates immediate local alerts and sends real-time notifications to improve response time and enhance safety.

2) PROBLEM OVERVIEW

- Conventional detectors often rely only on local alarms.
- If no one is nearby, critical hazards may go unnoticed.
- Advanced smart systems can be costly and complex.
- There is a need for an affordable, reliable, and connected safety solution.

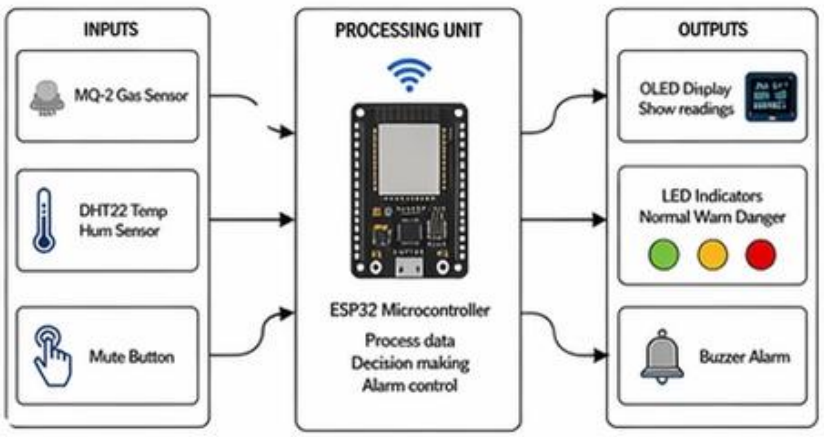
3) OBJECTIVES

- Integrate gas/smoke and temperature sensing in one system.
- Provide immediate audible and visual local alerts.
- Send real-time IoT notifications to smartphones.
- Deliver a low-cost and easy-to-install solution for homes and small businesses.

4) METHODOLOGY / SYSTEM DESIGN

- **Processing Unit:** ESP32 microcontroller with built-in Wi-Fi
- **Sensors:** MQ-2 Gas & Smoke Sensor, DHT22 Temperature & Humidity Sensor
- **Outputs:** 0.96-inch OLED Display, Green/Yellow/Red LEDs, Buzzer, Mute Button
- **IoT Platform:** Blynk for remote monitoring and notifications
- **Logic:** MQ-2 warm-up, baseline calibration, threshold-based decision making
- **System States:** NORMAL, WARNING, DANGER

System Architecture Diagram



System State Summary

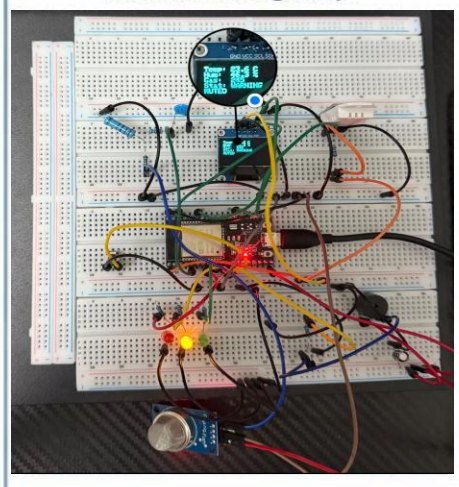
STATE	CONDITION	INDOCATION / ACTION
NORMAL	Safe values	Green LED
WARNING	Gas/temperature exceeds warning threshold	Yellow LED
DANGER	Gas/temperature exceeds danger threshold	Red LED + Buzzer + IoT Alert

5) TOOLS / HARDWARE COMPONENTS

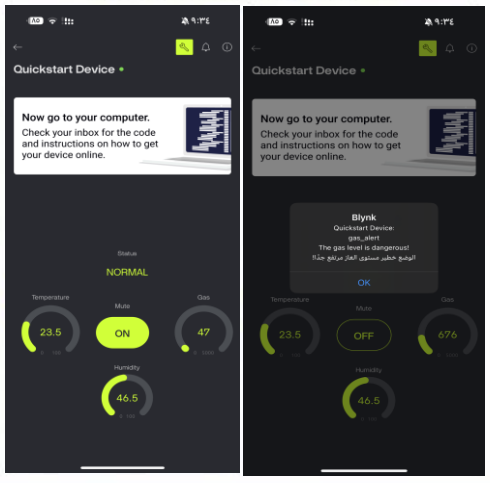
- ESP32
- MQ-2 Gas and Smoke Sensor
- DHT22 Sensor
- OLED Display
- Buzzer
- LED Indicators
- Push Button
- Blynk IoT
- Arduino IDE

6) RESULTS AND DISCUSSION

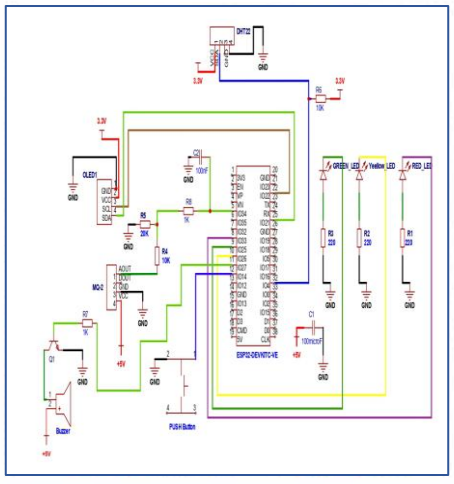
Hardware Testing Setup



Blynk IoT Monitoring Interface



Circuit Diagram



- The prototype successfully monitored gas level, temperature, and humidity in real time.
- The OLED displayed live sensor readings and system status.
- The system correctly transitioned between Normal, Warning, and Danger states.
- In danger conditions, the buzzer and red LED were activated, and Blynk sent remote alerts.
- The experimental setup demonstrated practical, stable, and responsive performance.

7) CONCLUSION

The proposed system provides a practical, low-cost, and scalable solution for early gas leak and fire detection. By combining continuous sensing, local alarms, and IoT-based notifications, it reduces detection-to-response time and improves safety for homes, restaurants, workshops, and other small environments.

Homes · Restaurants · Workshops · Small Businesses

Keywords: IoT, ESP32, Gas Leak Detection, Fire Detection, Blynk, Smart Safety System